Write a paper on the improvement of mechanical properties, dielectric properties and aging resistance of NBR rubber materials using nano-TiO2

Enhancement of Mechanical Properties, Dielectric Properties, and Aging Resistance of NBR Rubber Using Nano-TiO_2

Abstract

Nitrile butadiene rubber (NBR) is a polar elastomer widely used in industrial applications requiring oil resistance and a broad service temperature range. However, NBR does not exhibit self-reinforcing behavior due to its irregular random copolymer structure, resulting in comparatively weak mechanical properties. The incorporation of titanium dioxide (TiO_2) nanoparticles has emerged as an effective strategy for simultaneously enhancing the mechanical properties, dielectric properties, and aging resistance of NBR composites. This paper reviews the fundamental principles governing these enhancements, including the mechanisms of filler dispersion, interfacial interactions, crosslinking behavior, and the influence of filler loading. The analysis demonstrates that nano-TiO_2, through its dual role as a reinforcing filler and functional additive, can significantly improve the performance of NBR for demanding applications in dielectric elastomer actuators and high-performance rubber products.

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1. Introduction

Acrylonitrile butadiene rubber (NBR) is a copolymer of acrylonitrile and butadiene that exhibits beneficial properties including a broad temperature application range, excellent oil and solvent resistance, and relatively low cost . Due to acrylonitrile having a polar functional group (C≡N), NBR presents a specialty polar rubber with a high dielectric constant, making it attractive for dielectric elastomer actuator (DEA) applications . However, NBR does not exhibit self-reinforcing behaviors, as

there is no strain-induced crystallization due to its irregular molecular structure of random copolymers, resulting in comparatively weak mechanical properties .

Titanium dioxide (TiO_2) nanoparticles have attracted considerable attention due to their unique characteristics, including chemical stability, non-toxicity, high reactivity, biocompatibility, and low cost . TiO_2 has been widely employed as a white pigment in rubber industries, but recent research has demonstrated its potential as an effective reinforcing filler that can significantly improve the mechanical, dielectric, and aging properties of NBR composites . This paper examines the fundamental principles by which nano-TiO_2 enhances these properties, focusing on the mechanisms of reinforcement and the critical role of filler dispersion and interfacial interactions.

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2. Enhancement of Mechanical Properties

2.1 Cure Characteristics and Crosslink Density

The incorporation of TiO_2 into NBR significantly influences the vulcanization behavior. The addition of TiO_2 reduces scorch time (t_{s1}) and optimum cure time (t_{90}) because TiO_2 metal oxide acts as a co-activator stimulating the crosslinking during sulfur curing . The cure rate index (CRI) increases with TiO_2 content up to 30 phr and then remains constant, supporting that TiO_2 activates faster curing in TiO_2-filled NBR composites compared to the unfilled compound .

The torque difference ($M_h - M_l$) indirectly reflects the crosslink density in rubber vulcanizates. $M_h - M_l$ increases with the addition of TiO_2, indicating a higher crosslink density caused by strong filler-rubber interactions. These physical interactions occur between polar -C≡N groups in NBR and -OH groups on TiO_2 surfaces, including both terminal OH and bridge OH groups generated by the reaction of surface oxide with moisture in the air . The proposed physical interaction is shown schematically in the literature, where hydrogen bonds form between the TiO_2 surface hydroxyl groups and the NBR acrylonitrile functional groups .

2.2 Tensile Strength and Elongation at Break

The stress-strain behavior of TiO_2-filled NBR composites shows significant improvement over pure NBR. Pure NBR exhibits the lowest stress-strain curve with elongation at break approximately 550%, and without self-reinforcing behavior due to its irregular structure . The stress steadily increases with applied strain and then escalates sharply for TiO_2-filled NBR composites, particularly at 90 phr TiO_2 content. The strain at the onset of the stress upturn corresponds to the onset of crystallization. In the presence of TiO_2, stronger chain alignment in the strain direction is caused by the additional crosslinks created by filler-rubber interactions .

**Table 1: Mechanical Properties of TiO_2-Filled NBR Composites with Various TiO_2 Loadings **

Sample	Tensile Strength (MPa)	Elongation at Break (%)	M100 (MPa)	M300 (MPa)	Hardness (Shore A)
Pure NBR	2.82 ± 0.11	550 ± 15	0.71 ± 0.01	1.20 ± 0.04	35
NBR/TiO_2 30 phr	3.77 ± 0.16	565 ± 15	1.01 ± 0.07	2.08 ± 0.05	45
NBR/TiO_2 70 phr	4.16 ± 0.18	621 ± 14	1.21 ± 0.02	2.59 ± 0.05	48
NBR/TiO_2 90 phr	5.68 ± 0.13	768 ± 11	1.45 ± 0.01	2.79 ± 0.04	49
NBR/TiO_2 110 phr	5.24 ± 0.18	699 ± 19	1.50 ± 0.03	2.87 ± 0.03	51

The optimal TiO_2 content for maximum tensile strength and elongation at break is 90 phr. Tensile strength and elongation at break are increased by **144.8% and 40.1%**, respectively, over pure NBR . This significant improvement is attributed to good filler-rubber interactions, which result in good filler dispersion in the NBR matrix and efficient stress transfer from matrix to filler . On further increasing TiO_2 content to 110 phr, the tensile strength and elongation at break slightly decrease, probably due to agglomeration of TiO_2 in the NBR matrix, which degrades filler-rubber interactions and impedes stress transfer .

2.3 Modulus and Hardness

The moduli at 100% and 300% elongations are indirectly correlated to stiffness and rigidity. Moduli at 100% and 300% elongations increase with TiO_2 content, reaching 111% and 139% increases, respectively, for NBR composites with 110 phr TiO_2 content . This indicates that the stiffness of the composites has increased due to crosslinking, in agreement with M_h and $M_h - M_l$ results .

The reinforcing index, defined as the ratio of M300 to M100, is higher for TiO_2-filled NBR composites than for pure NBR vulcanizate, confirming that TiO_2 acts as a reinforcing filler in NBR rubber. The 70 phr TiO_2-filled composites exhibit the highest reinforcement index .

Hardness of NBR composites is also affected by TiO_2 content, with an increase of approximately 46% at 110 phr TiO_2 loading compared to pure NBR. This is explained by the increased crosslink density and the rigid inorganic TiO_2 particles being noticeably stiffer than the rubber matrix .

2.4 Dynamic Mechanical Properties

Dynamic mechanical thermal analysis (DMTA) reveals that the addition of TiO_2 filler gives a higher storage modulus, a shift in glass transition temperature (Tg) to a higher temperature, and reduced mechanical damping . The storage modulus relates to stiffness, while tan δ is the ratio of loss modulus to storage modulus, relating to molecular mobility and phase transition .

The shift in Tg to higher temperatures indicates restricted segmental motion of the polymer chains due to physical interactions between polymer chains and filler surfaces, increased crosslink density, and formation of bound rubber layers. The reduced damping (tan δ) is consistent with the formation of a more constrained polymer network with reduced energy dissipation during cyclic deformation.

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3. Enhancement of Dielectric Properties

3.1 Dielectric Constant Enhancement

The addition of TiO_2 to NBR rubber increases the dielectric constant of the resultant composites in the tested frequency range from 10^2 to 10^5 Hz . The dielectric constant increases with TiO_2 content because TiO_2 has a higher dielectric constant than pure NBR (TiO_2 approximately 48-100, NBR approximately 12-18 at 10^2 Hz) .

The maximum dielectric constant of pure NBR, approximately 18.83 at 10^2 Hz, is attributed to strong contributions from interfacial and dipolar polarization in the low-frequency range. The high dielectric constant of pure NBR is due to the orientation polarization of the -C≡N dipoles. On increasing the frequency to 10^5 Hz, the dielectric constant decreases because the dipoles are incapable of reorienting rapidly enough to follow the applied frequency .

**Table 2: Dielectric Properties of TiO_2-Filled NBR Composites **

Sample	Dielectric Constant (10^2 Hz)	Dielectric Loss (1 kHz)
Pure NBR	~18.83	0.0137
30 phr TiO_2	~20.0 (estimated)	0.0132
110 phr TiO_2	20.28	Not reported

The 110 phr TiO_2-filled composites show the highest dielectric constant of 20.28 at a frequency of 10^2 Hz . The increased dielectric constant is partly attributed to better dipole orientation polarization due to the segment arrangement of rubber chains. After the incorporation of TiO_2 filler, the composites have more interfaces enabling interfacial polarization . The presence of TiO_2 in the composites increases the dielectric constant through contributions from both the filler's intrinsic dielectric constant and Maxwell-Wagner interfacial polarization at the filler-matrix interfaces.

3.2 Surface Modification Effects

Surface modification of TiO_2 nanoparticles can significantly enhance dielectric induction efficiency. Polydopamine (PDA) coating of TiO_2 nanoparticles improves compatibility between the filler and the NBR matrix, leading to better filler dispersion and reduced dielectric loss .

**Table 3: Effect of Surface Modification on Dielectric and Electromechanical Properties **

Sample	Elastic Modulus (MPa)	Dielectric Constant (1 kHz)	Dielectric Loss (1 kHz)	Max Strain (%)	Breakdown Strength (kV/mm)
0 phr	1.38	13.43	0.0137	7.58	45
10 phr TiO_2	1.37	14.38	0.0127	11.38	47
20 phr TiO_2	1.46	15.18	0.0130	11.51	57
30 phr TiO_2	1.52	15.53	0.0132	12.31	63
20 phr TiO_2-(PDA+KH570)	1.43	15.26	0.0126	15.77	60

Combined covalent and noncovalent modification using PDA followed by silane coupling agents (KH570) demonstrates even better performance. The TiO_2-(PDA+KH570)/NBR composites exhibit dielectric constants of 14.55–15.73 at 1 kHz, higher than both pure NBR (13.43) and unmodified TiO_2/NBR composites . The actuated strain reaches 15.77% at 20 phr loading, representing more than a 100% increase over pure NBR .

3.3 Breakdown Strength Enhancement

TiO_2 nanoparticles can enhance the dielectric breakdown strength of NBR composites. The breakdown strength of pure NBR is 45 kV/mm, while TiO_2-filled NBR composites achieve higher breakdown strengths up to 65 kV/mm with modified TiO_2 at 30 phr loading . The mechanism of breakdown strength enhancement is attributed to the ability of nanoparticles to prevent charges from moving freely through the insulating material. TiO_2 nanoparticles diffuse into the voids and gaps within the NBR matrix, forming new strong bonds and filling the voids. The nanoparticles release free electrons and stop their movement under the applied voltage, preventing collisions with internal molecules that would generate new free electrons and lead to breakdown .

TiO_2 nanoparticles exhibit higher breakdown strength enhancement than MgO nanoparticles, primarily due to TiO_2's higher dielectric constant and better dispersion in the NBR matrix . The AC breakdown

voltage of NBR composites loaded with TiO_2 nanoparticles reaches values significantly higher than those of MgO-loaded composites at all weight fractions tested .

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4. Aging Resistance Enhancement

4.1 Thermal Aging Resistance

Thermal aging has been investigated for nano-TiO_2 filled rubber composites. Research on nano-TiO_2 filled NBR composites shows that thermal aging has no significant influence on the antimicrobial properties of the composites . The thermal aging does not affect the TiO_2's ability to function as an antimicrobial agent, indicating that the TiO_2 particles remain stable within the rubber matrix even after aging .

The presence of TiO_2 in NBR composites can contribute to improved aging resistance through several mechanisms. TiO_2 acts as a UV absorber and can prevent photo-oxidative degradation of the rubber matrix . The strong filler-rubber interactions via hydrogen bonding between hydroxyl on the TiO_2 surface and acrylonitrile-functional groups create a more stable composite structure that is less susceptible to thermal-oxidative degradation .

4.2 Hot Air Aging Studies

Studies on hot air aging of polymer nanocomposites containing nano-TiO_2 have been performed . The effect of increasing nano-fillers on mechanical properties like tensile strength, modulus, and elongation at break was studied after hot air aging. The incorporation of nano-TiO_2 helps maintain mechanical properties after aging due to the stabilizing effect of the nanoparticles .

4.3 Antimicrobial Properties as Aging Indicator

Nano-TiO_2 adds antimicrobial functionality to NBR composites, which can serve as an indicator of filler stability after aging. The antimicrobial properties of TiO_2/NR and TiO_2/NBR composites are greatly improved compared to unfilled composites, reaching a high level when the loading of TiO_2 is over 2 phr . The thermal aging has no influence on the antimicrobial properties of the TiO_2-filled composites, demonstrating that the TiO_2 nanoparticles maintain their functionality even after exposure to aging conditions .

The antimicrobial mechanism of TiO_2 involves the generation of reactive oxygen species (ROS) under UV light, which can damage bacterial cell walls. The stability of this function after aging indicates that the TiO_2 particles remain photoactive and well-dispersed in the rubber matrix .

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5. Mechanisms of Property Enhancement

5.1 The Role of Filler Dispersion

The quality of TiO_2 dispersion in the NBR matrix is critical for achieving optimal mechanical, dielectric, and aging properties. Studies on nano-TiO_2 filled rubber composites have shown that B-TiO_2 (20-40 nm) exhibits good dispersion in NBR, with aggregate sizes predominantly less than 100 nm . Good dispersion ensures:

- Maximum surface area for interfacial interactions
- Uniform distribution of reinforcing effects
- Consistent crosslink density throughout the composite
- Effective stress transfer from matrix to filler

SEM micrographs of TiO_2-filled NBR composites show that incorporating TiO_2 in the NBR matrix induces rougher fracture surfaces, so more energy is needed to break the sample . The strong hydrogen bonds between TiO_2 particles and the NBR matrix can prevent the formation of defects that initiate or propagate fractures .

5.2 Filler-Rubber Interactions

The polar nature of NBR, imparted by the acrylonitrile content, creates favorable conditions for interaction with the surface of TiO_2 nanoparticles. The surface of TiO_2 particles contains hydroxyl groups (-OH) that can form physical interactions with the polar -C≡N groups of NBR . These interactions contribute to:

- **Improved filler dispersion**: Well-dispersed TiO_2 particles provide maximum surface area for reinforcement and dielectric enhancement

- **Strong filler-rubber adhesion**: Strong interfacial interactions enable effective transfer of stress and electric field from the rubber matrix to the TiO_2 particles

- **Enhanced crosslink density**: The physical interactions create additional crosslinks that complement the chemical crosslinks formed during vulcanization

- **Bound rubber formation**: The immobilized polymer layers effectively increase the volume fraction of the reinforcing phase

5.3 Synergistic Enhancement

The unique advantage of TiO_2/NBR composites is the simultaneous improvement of mechanical, dielectric, and aging properties. This synergistic enhancement occurs through:

Mechanical Enhancement: The addition of TiO_2 improves tensile strength (up to 144.8% increase at 90 phr), elongation at break (up to 40.1% increase), and hardness (up to 46% increase) through filler-rubber interactions, stress transfer, and increased crosslink density .

Dielectric Enhancement: The addition of TiO_2 increases the dielectric constant (up to 20.28 at 10^2 Hz) through filler contribution, interfacial polarization, and improved dipole orientation polarization .

Aging Resistance: The presence of TiO_2 provides UV absorption and thermal stabilization, maintaining antimicrobial functionality even after aging .

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6. Conclusions

The enhancement of mechanical properties, dielectric properties, and aging resistance in NBR rubber by nano-TiO_2 is governed by several interrelated principles:

1. **Mechanical Properties**: TiO_2 acts as a reinforcing filler, increasing crosslink density through strong filler-rubber interactions between polar -C≡N groups in NBR and -OH groups on TiO_2 surfaces. The optimal TiO_2 content for maximum tensile strength and elongation at break is 90 phr, resulting in a 144.8% increase in tensile strength and a 40.1% increase in elongation at break .

2. **Dielectric Properties**: TiO_2 nanoparticles with high dielectric constant increase the effective dielectric constant of the composite through filler contribution and interfacial polarization. The 110 phr TiO_2-filled composites show the highest dielectric constant of 20.28 at 10^2 Hz, compared to approximately 18.83 for pure NBR .

3. **Breakdown Strength**: TiO_2 nanoparticles enhance dielectric breakdown strength by filling voids, forming strong bonds, and immobilizing free electrons. Modified TiO_2 composites achieve breakdown strengths up to 65 kV/mm .

4. **Aging Resistance**: TiO_2 provides UV absorption and thermal stabilization, maintaining antimicrobial functionality even after aging. Thermal aging has no significant influence on the antimicrobial properties of TiO_2-filled NBR composites .

5. **Surface Modification**: Bio-inspired polydopamine coating and silane coupling agents improve filler dispersion, reduce dielectric loss, and enhance breakdown strength, leading to actuated strains as high as 15.77% (20 phr modified TiO_2), representing more than a 100% increase over pure NBR .

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